

Market Microstructure Frameworks for Spoofing Detection: Isolating Manipulative Intent from Legitimate Liquidity Provision

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Abstract

In this paper we detect the bad guys and we find how to tell them they are actually bad guys. Hopefully, we will be able to do this without accusing the good guys of being bad guys.

Keywords. Spoofing detection; market manipulation; limit order book; high-frequency trading; anomaly detection; market maker; market microstructure

1 Introduction

The transition of global financial exchanges to fully electronic, continuous double auction matching engines has fundamentally transformed the mechanics of price discovery and liquidity provision. In modern financial ecosystems, the Limit Order Book (LOB) serves as the central nervous system of the market, aggregating the continuous stream of bid and ask limit orders submitted by diverse market participants. While the advent of algorithmic and high-frequency trading (HFT) has undeniably narrowed bid-ask spreads and enhanced short-term market efficiency, it has concurrently introduced profound vulnerabilities. The microsecond latency environment allows sophisticated algorithms to exploit the structural mechanics of the LOB, executing complex manipulative strategies that deceive heuristic-based trading algorithms and human traders alike. Among these manipulative practices, spoofing represents one of the most pervasive and challenging threats to market integrity. Spoofing is a deceptive algorithmic strategy wherein a market participant places a massive, non-bona fide order on one side of the LOB with the explicit intent of canceling it before execution. The presence of this immense, artificial volume creates a false impression of directional market pressure. This artificial pressure manipulates the micro-price and induces other market participants—particularly algorithms that rely on order book imbalance indicators—to adjust their quotes or submit market orders. Once the price has been artificially driven to a favorable level, the spoofer executes a smaller, bona fide

order on the opposite side of the book and immediately withdraws the massive deceptive orders, thereby profiting from the artificially induced price dislocation. The European Securities and Markets Authority (ESMA), through the Market Abuse Regulation (MAR) [3], strictly prohibits such practices, categorizing them as behaviors that disrupt the functioning of the trading system and create misleading signals regarding the supply and demand of financial instruments. For regulatory and supervisory authorities such as the Commissione Nazionale per le Società e la Borsa (CONSOB), the detection and prosecution of spoofing present a severe analytical challenge. The physical actions comprising a spoofing strategy—posting limit orders, updating quotes, and rapidly canceling orders—are mechanically indistinguishable from the legitimate actions of a bona fide market maker providing liquidity. Both spoofers and market makers exhibit exceedingly high order-to-trade ratios (OTTR) and operate at high frequencies. Traditional market surveillance mechanisms, which historically relied on static heuristic thresholds, volumetric alerts, or simple cancellation ratios, are fundamentally inadequate in this environment. These legacy systems generate an overwhelming volume of false positives, frequently flagging legitimate liquidity provision and inventory management as manipulative market abuse.

2 The Microstructural Divergence: Market Making Versus Spoofing

To construct a mathematical model capable of detecting market manipulation, it is first necessary to rigorously define the divergent objective functions that govern legitimate market making and spoofing. While both participants generate massive volumes of limit order messages and cancellations, their underlying economic motivations, risk profiles, and optimizations are fundamentally opposed.

2.1 The Market Maker’s Problem

A market maker functions as a continuous liquidity provider, simultaneously quoting resting limit orders at the best bid and the best ask. The primary objective function of a market maker is to maximize the utility of gains derived from capturing the bid-ask spread across a vast number of transactions, while concurrently minimizing the variance of their portfolio and strictly managing inventory risk. In market microstructure theory, the pricing dynamics of a liquidity provider are governed by the interplay between the arrival rate of market orders and the accumulation of inventory.

The optimization problem for a market maker is frequently modeled using stochastic control techniques and Hamilton-Jacobi-Bellman (HJB) equations, where the dealer sets optimal quotes to balance the asymmetric arrival of buy and sell orders. ¹ If a market maker accumulates a long position (positive inventory), they are exposed to the risk of the underlying asset depreciating. To mitigate this inventory risk, the rational market maker will lower their quotes downwards—lowering their ask price to incentivize incoming buyers, and lowering their bid price to dissuade incoming sellers. ² The bid and ask prices are dynamically adjusted functions of the mid-price, the base spread required for profitability, and an inventory penalty parameter that increases as the dealer’s position deviates from neutrality.

Consequently, a market maker’s high cancellation rate is entirely reactive and protective. Limit orders are rapidly canceled and re-posted to reflect minute changes in the fundamental mid-price, to defend against adverse selection informed traders, and to recalibrate inventory penalty. The massive volume of orders and cancellations produced by a market maker is therefore characterized by symmetry across the LOB and a continuous presence at the top of the book to execute trades. Their expected directional gain from posting an order is near zero, as their profit is entirely derived from the spread rather than from inducing a directional shift in the mid-price.

2.2 The Spoofer’s Problem

In stark contrast, a spoofing agent does not seek to capture the bid-ask spread, nor do they operate as symmetric liquidity provider. A spoofer’s objective function is purely directional. The spoofer desires to execute a relatively small, bona fide order at a highly favorable price by artificially manipulating the micro-price dynamics of the LOB. To execute this strategy, the spoofer deploys a massive non-bona fide order on the opposite side of the LOB. This deliberate injection of spurious volume creates an extreme unbalanced environment, artificially inflating the visible depth of the book. In modern electronic markets, countless algorithmic traders utilize strategies that weight the volume at the bid against the volume at the ask to predict short-term price distributions. By overwhelming one side of the book, the spoofer pollutes these volume imbalance metrics, causing HBL algorithms to mathematically deduce an impending price shift and aggressively alter their own quoting behavior.

The critical mathematical dilemma for the spoofer lies in the placement of the non-bona fide order. The spoofer must balance the expected market manipulation gain against the explicit risk of unintended execution. If the spoofer places the massive non-bona fide order exactly at the top of the book (the best bid or ask), they generate the maximum psychological and algorithmic impact on the micro-price. However, doing so exposes the spoofer to massive adverse selection and inventory risk; an incoming aggressive market order could inadvertently execute against the spoofer’s massive fake order, leaving the spoofer holding an enormous, unwanted position. To optimize this expected cost function, sophisticated spoofers rarely post their deceptive orders at the absolute best price. Instead, they optimize the posting distance (measured in ticks or basis points) deeper within the book. By placing the order at Level 2, Level 3, or beyond, the spoofer ensures the volume is still visible to the matching engine’s data feed—and thus factored into the volume imbalance metrics of victim algorithms—while maintaining a structural buffer against immediate execution. Once the intended directional shift occurs and the spoofer’s small bona fide order is executed, the massive non-bona fide order is instantly canceled. The spoofer’s cancellation intent is not to manage inventory, but to terminate the manipulative episode the exact millisecond the cost optimization concludes.

3 Existing Approaches

The literature on spoofing detection is still relatively small, but it has moved from transparent microstructure rules toward sequence and graph models that try to capture richer strategic

Table 1: Core Differentiators for Surveillance Models

Microstructure Metric	Legitimate Market Maker	Manipulative Spoofing Agent
Primary Objective Function	Maximize bid–ask spread capture while minimizing inventory variance.	Minimize execution cost of a directional trade via artificial micro-price impact.
Order Book Imbalance	Generally symmetric, adapting quotes to maintain a neutral inventory.	Extremely asymmetric, deliberately creating massive, sudden imbalances (unbalanced quoting).
Cancellation Motivation	Reactive: mitigating adverse selection and updating quotes to track fair value.	Proactive: terminating the deceptive posture immediately after a targeted execution.
Posting Distance Optimization	Top of the book (Level 1) to maximize the probability of spread execution.	Deeper in the book (Level 2/3) to maximize visibility while minimizing execution risk.
Expected Directional Gain	Zero or negative (willing to pay the spread to lay off excessive inventory risk).	Strictly positive (the massive order statistically subsidizes the execution of the small order).

behavior. For the purpose of this paper, the most useful distinction is not simply between “statistical” and “machine-learning” models. It is between methods that identify a stylized spoofing episode from order-book mechanics, methods that monitor abnormal order-flow states, and methods that learn temporal or relational patterns from proprietary event data. [Figure 1](#) summarizes this taxonomy.

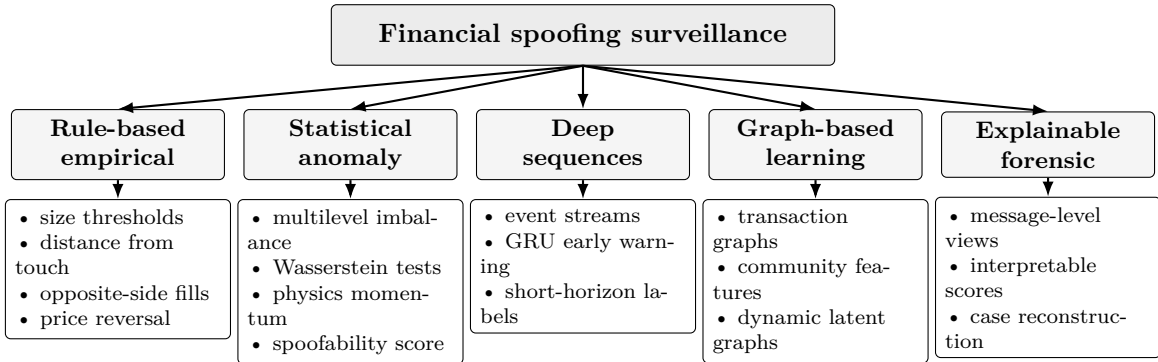


Figure 1: Method families in the spoofing-detection literature. The categories are not mutually exclusive, but they highlight the main modeling choice made by each strand of work.

Early empirical papers define spoofing through economically interpretable order-book conditions. The seminal study by [7] identifies large orders placed far from the best quote, followed by opposite-side trading and subsequent withdrawal, using account-level data from the Korea Exchange. Similar surveillance logic appears in Brazilian equities [9], FX spot markets [11],

cross-market FX settings [10], and Taiwan derivatives [1]. These studies are valuable because the alert logic can be explained to supervisors: suspicious activity is tied to order size, posting distance, cancellation timing, opposite-side execution, and short-lived price impact. Their main limitation is that thresholds are venue-specific and often depend on privileged message data or account identifiers.

A second line develops statistical and anomaly-oriented monitoring. [12] derive spoofing incentives from multilevel order-book imbalance and propose Wasserstein-distance monitoring for real-time detection. [8] use a statistical-physics representation of the order book to detect abnormal momentum in crypto markets. More recent explainable approaches also fall partly in this family: forensic visualization reconstructs message-level spoofing episodes in prosecuted cases [2], while interpretable probabilistic neural networks estimate the conditional gain from manipulating a centralized crypto order book [4]. These methods improve on pure threshold rules by modeling the state of the book, but many papers still report limited benchmark metrics or rely on non-public datasets.

Machine-learning work mainly adds temporal or relational structure. [13] formulate spoofing as an early-warning problem and train a GRU model on cryptocurrency order-book event streams. Graph-based methods then move the unit of analysis from isolated orders to networks of traders and transactions, which is important for detecting coordinated or “conspiracy” spoofing [5, 6, 14]. The strongest reported graph models combine temporal transaction graphs, community-level features, and dynamic latent representations, but their practical value is difficult to assess without open labels and chronological cross-market validation.

Overall, the literature suggests two requirements for a robust surveillance model. First, the features should remain microstructure-grounded: a detector should know whether suspicious volume is large, distant from the touch, rapidly cancelled, and conditional on an opposite-side execution. Second, the evaluation should respect the enforcement setting: labels are noisy, true positives are rare, and false positives impose substantial analyst costs. This motivates the agent-level, spatio-temporal framework developed below, which keeps the interpretability of rule-based surveillance while using the full order-flow sequence available in proprietary data.

4 Data

5 Model

In this section we provide the base reasoning and metrics used to distinguish between legitimate market making behaviour and possible spoofing strategies.

5.1 A Spatio-Temporal Framework for Spoofing Detection

To rigorously distinguish manipulative spoofing from legitimate liquidity provision, we must define the microstructural signature of the strategy not merely as a spatial anomaly within the Limit Order Book (LOB), but as a spatio-temporal sequence of conditional events.

We formalize the execution timeline of a spoofing agent i into three distinct states:

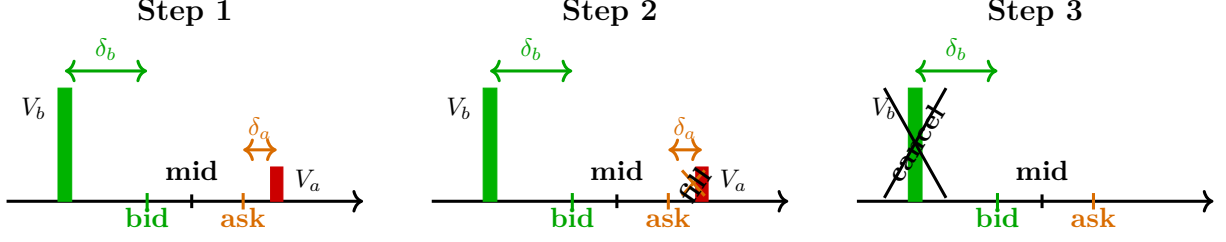


Figure 2: Timeline of a spoofing episode

1. **State 1 (Deceptive Posturing):** The agent posts a massive, non-bona fide order (V_b) deep in the book at distance δ_b from the same-side best quote, simultaneously posting a small, bona fide order (V_a) at or near the best opposite quote ($\delta_a \approx 0$).
2. **State 2 (Execution):** The artificial volume imbalance shifts the micro-price, resulting in the execution of the bona fide order V_a .
3. **State 3 (Risk Termination):** Immediately following the execution of V_a , the agent cancels the massive non-bona fide order V_b to neutralize execution risk.

To capture this dynamic, we define a set of metrics that isolate the agent’s spatial footprint and their conditional reactivity.

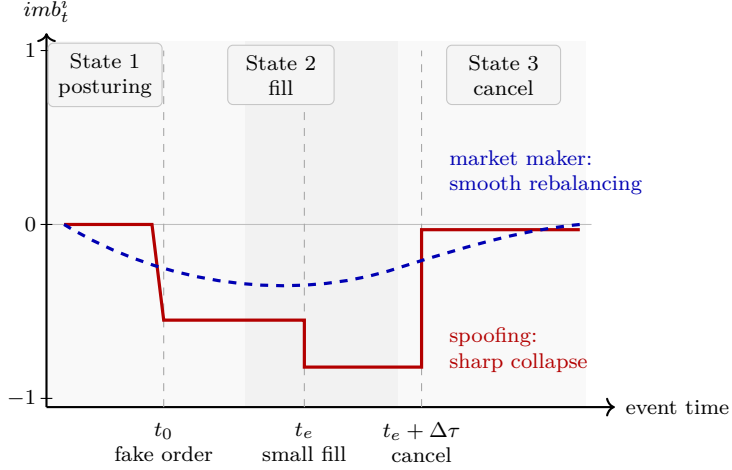
5.1.1 Normalized Spatial Imbalance

Traditional order book imbalance metrics often fail to accurately weight deep-book liquidity. While a spoofer seeks to maximize visible volume, the probability of execution—and thus the true risk borne by the agent—decays non-linearly as distance from the mid-price increases.

We explicitly assume that the manipulative impact of an order is a saturating function of its distance from the best quote. Let V_a and V_b represent the volumes of the agent’s ask and bid orders, respectively. We define the *Normalized Distance-Weighted Imbalance* for agent i at event-time τ as:

$$imb_{\tau}^i = \left. \frac{V_a(1 - e^{-\kappa\delta_a}) - V_b(1 - e^{-\kappa\delta_b})}{V_a + V_b} \right|_i \quad (1)$$

where $\kappa > 0$ is an asset-specific decay parameter calibrated to the instrument’s typical liquidity density and spread variance. The term $(1 - e^{-\kappa\delta})$ acts as an asymptotic filter, strictly bounding the distance multiplier within $[0, 1)$. This guarantees that $imb_{\tau}^i \in [-1, 1]$, providing a way to compare imbalance across different assets and time periods while emphasizing the strategic placement of orders.



For a buy-intent spoofing episode, the same pattern is mirrored above zero.

Figure 3: Stylized event-time behavior of the normalized distance-weighted imbalance. A spoofing episode produces a large imbalance and, after the small opposite-side execution, a sharp return toward zero when the non-bona fide order is cancelled. Inventory rebalancing by a market maker should instead evolve more smoothly.

5.1.2 Conditionality and Intent Isolation

A static imbalance metric is insufficient to isolate intent, as a legitimate market maker engaged in asymmetric inventory management may temporarily exhibit extreme imb_t^i values. The fundamental differentiator lies in the conditionality of State 3. For a market maker, the cancellation of a deep resting order is driven by adverse selection or mid-price momentum. For a spoofer, the cancellation is strictly conditional upon the execution of the opposite-side bona fide order.

Let e denote the specific execution event of the small order V_a at timestamp t_e . We define a microscopic time window $\Delta\tau$ bounding the mechanical latency of the strategy. The *Spoofing Conditionality Index (SCI)* measures the structural collapse of the agent's LOB presence immediately surrounding the fill:

$$SCI_e^i = |imb_{t_e^-}^i - imb_{t_e^+}^i| \quad (2)$$

where t_e^- is the state immediately preceding the fill, and $t_e^+ = t_e + \Delta\tau$. An extreme SCI_e^i value indicates the sudden evaporation of the opposite-side massive order, a mechanical signature of manipulative risk termination.

To aggregate this behavior over a trading session and filter out coincidental reactive cancellations by high-frequency market makers, we define the *Conditional Probability Score (CPS)* for agent i :

$$CPS^i = \frac{1}{N} \sum_{e=1}^N \mathbb{I}(SCI_e^i > \gamma) \quad (3)$$

where N is the total number of small-order executions for the agent, $\mathbb{I}$ is the indicator function, and γ is an empirically derived threshold representing a structural imbalance shock. A legitimate market maker's CPS^i will converge toward zero, whereas a systematic spoofing

algorithm will exhibit a CPS^i approaching unity.

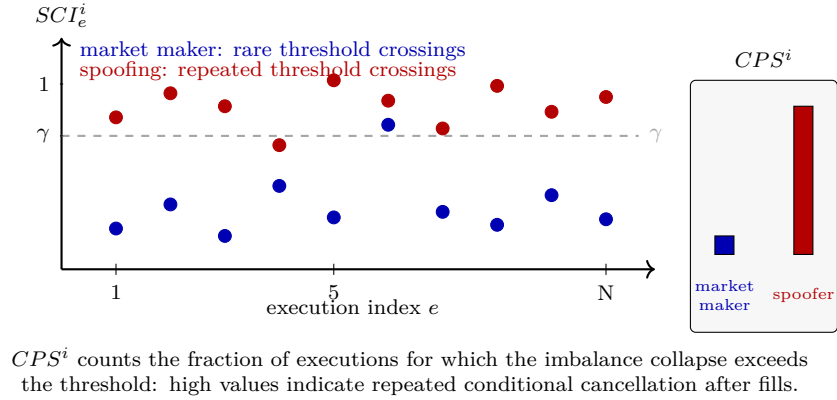


Figure 4: Operational interpretation of the Conditional Probability Score. Each point is one small-order execution for the agent. The score increases when many execution-level imbalance collapses SCI_e^i exceed the threshold γ .

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